

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

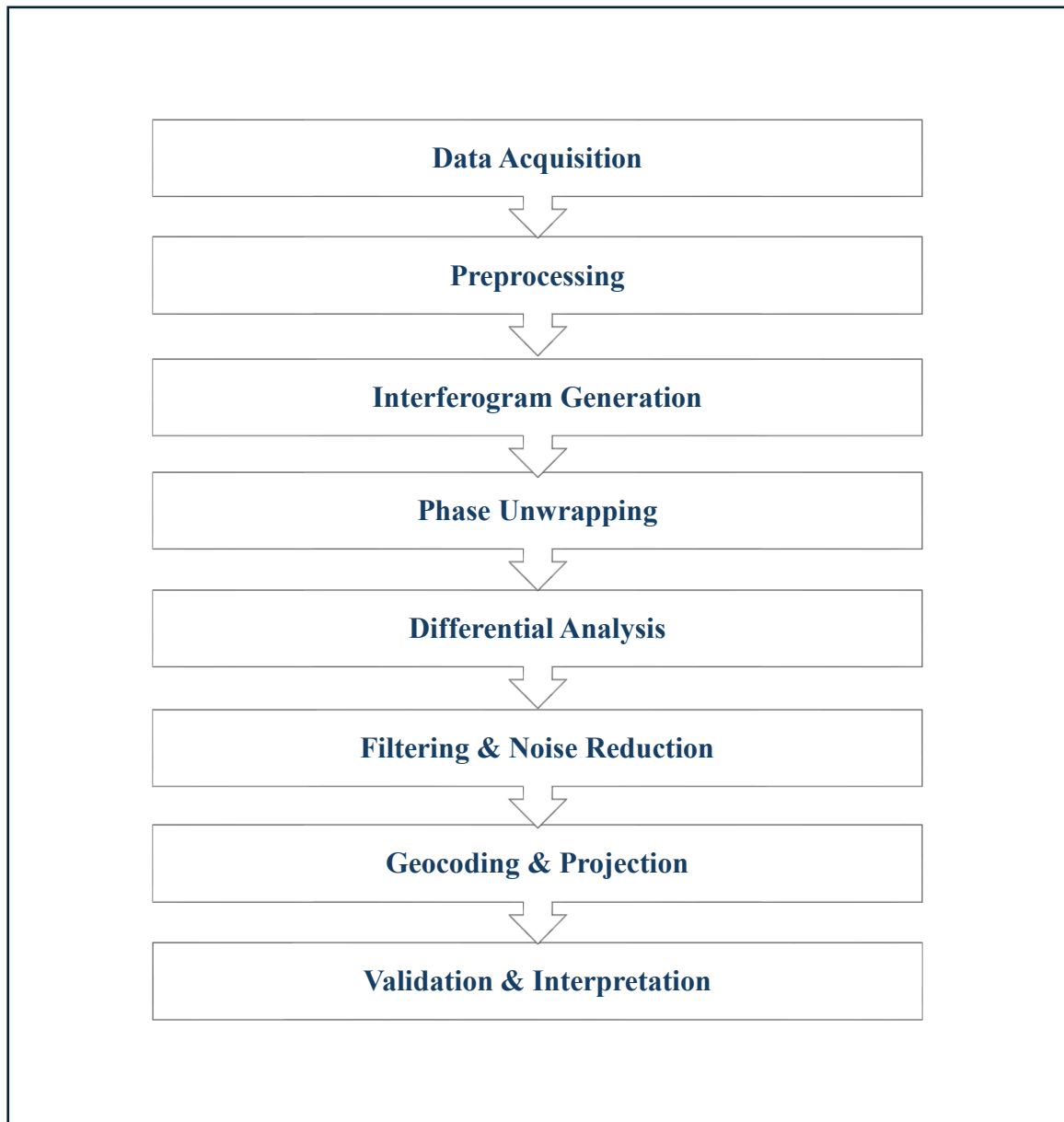
Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimeter-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 903/2

Bridge Chainage – 967+030

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MNB

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

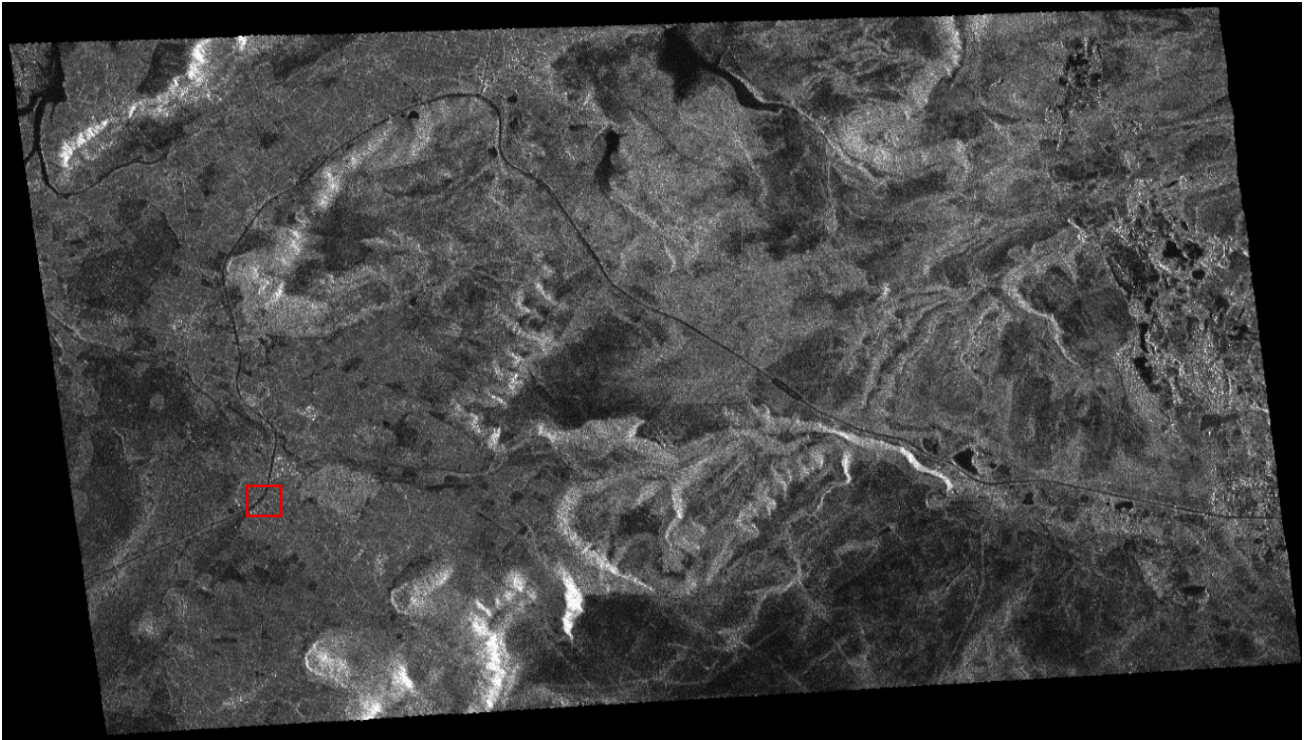
Flight Direction - DESCENDING

Polarization – VV+VH

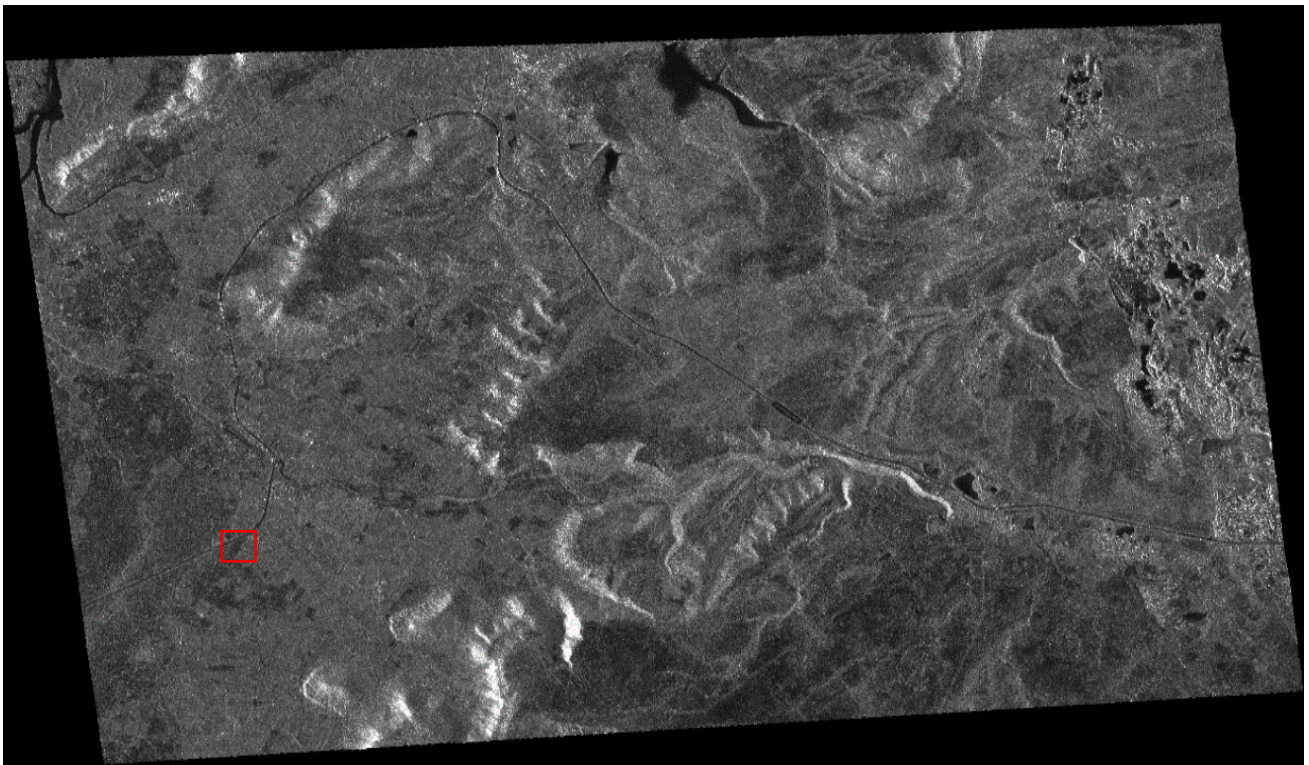
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

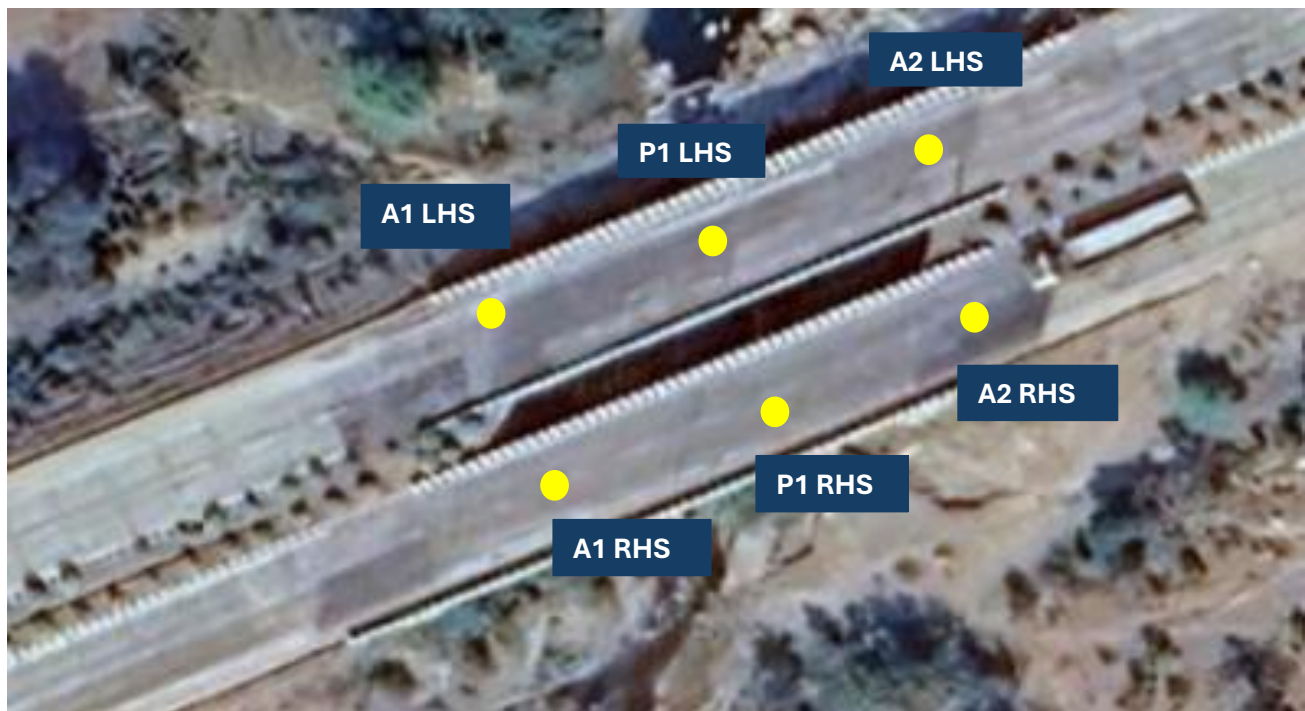


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

MNB – CH-967+030

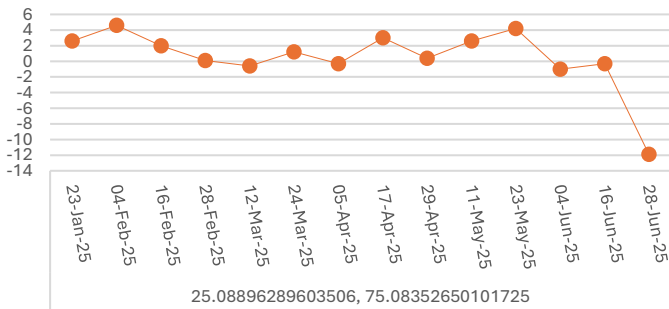
Length – 55.48M



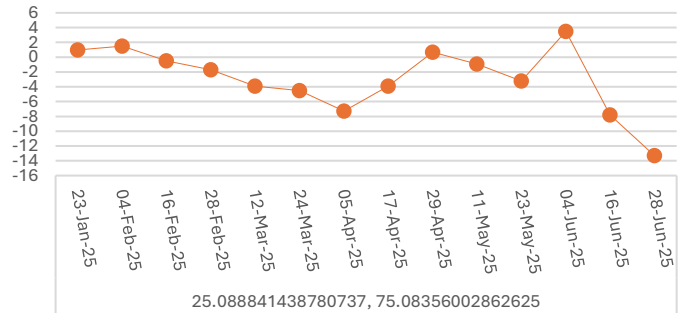
Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1	Abutment 1		
P1	Pier 1		
A2	Abutment 2		
LHS	Left Hand Side		
RHS	Right Hand Side		

3.1 VV Displacement from 23 Jan 2025 To 28 June 2025

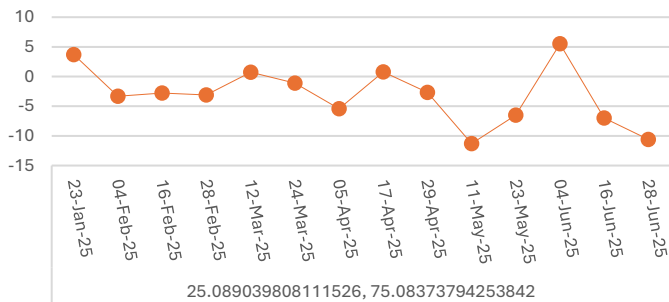
Displacement @ A1 LHS VV



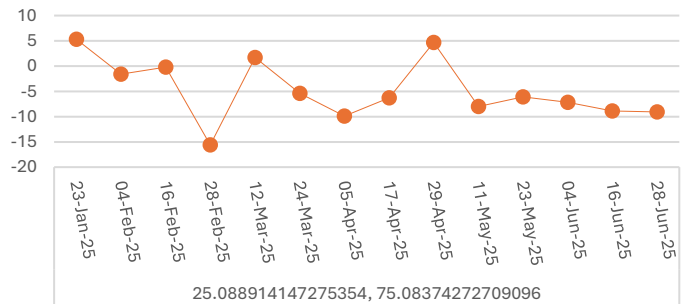
Displacement @ A1 RHS VV



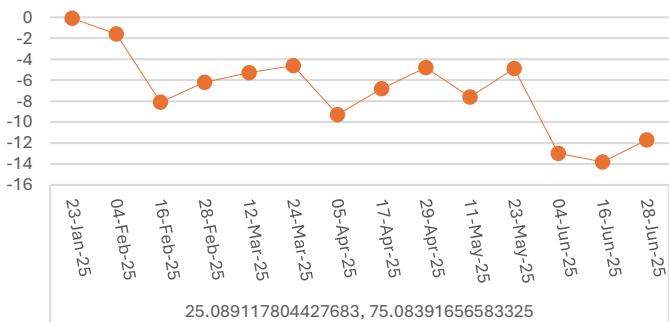
Displacement @ P1 LHS VV



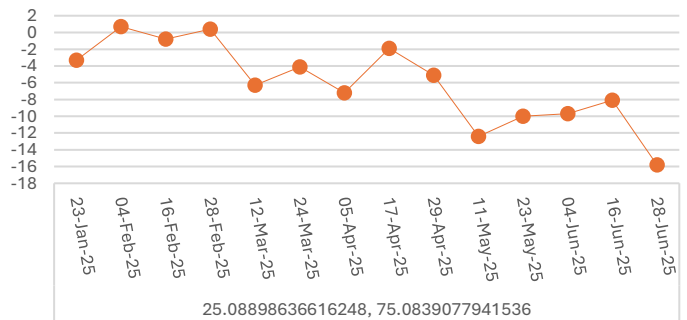
Displacement @ P1 RHS VV



Displacement @ A2 LHS VV

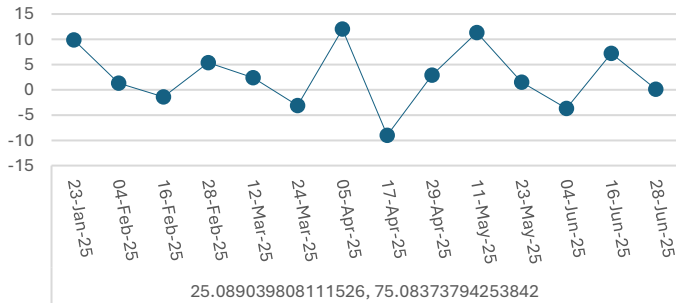


Displacement @ A2 RHS VV

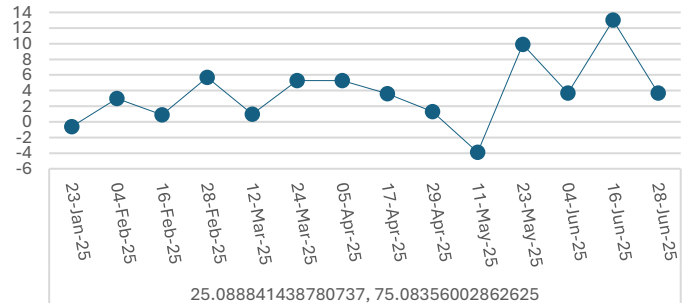


3.2 VH Displacement From 23 Jan 2025 To 28 June 2025

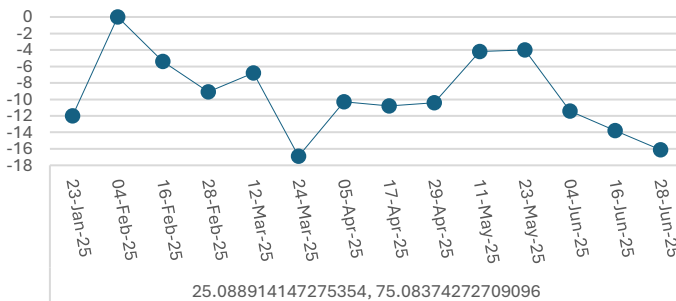
Displacement @ A1 LHS VH



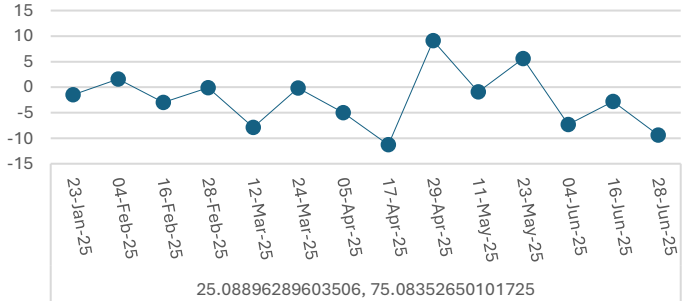
Displacement @ A1 RHS VH



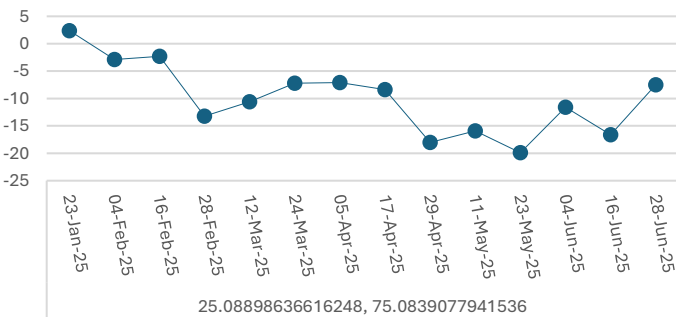
Displacement @ P1 LHS VH



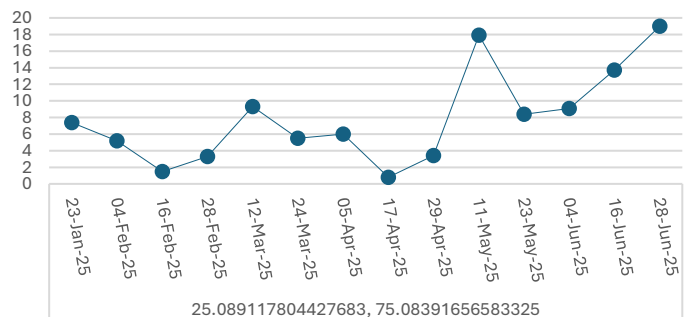
Displacement @ P1 RHS VH



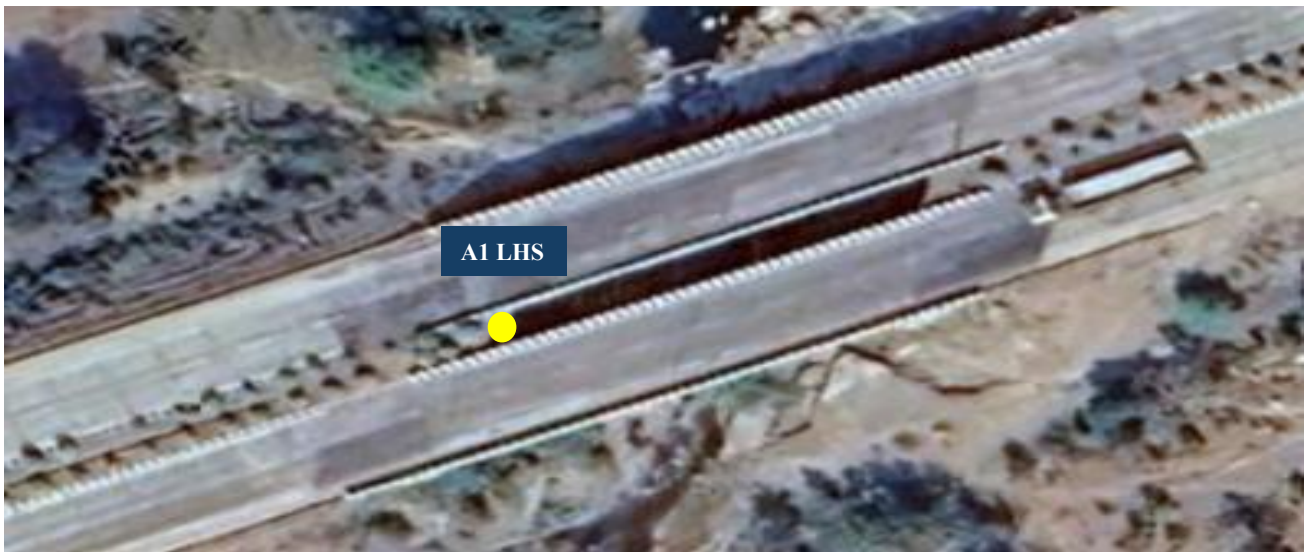
Displacement @ A2 LHS VH



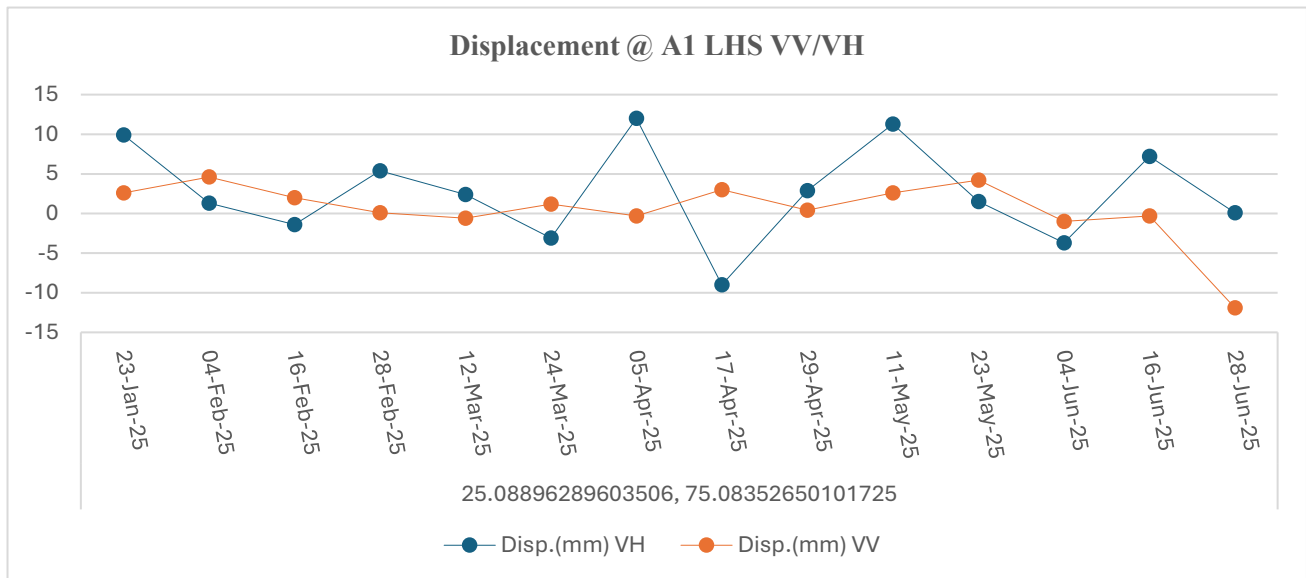
Displacement @ A2 RHS VH



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.08896289603506, 75.08352650101725)



Key Observations:

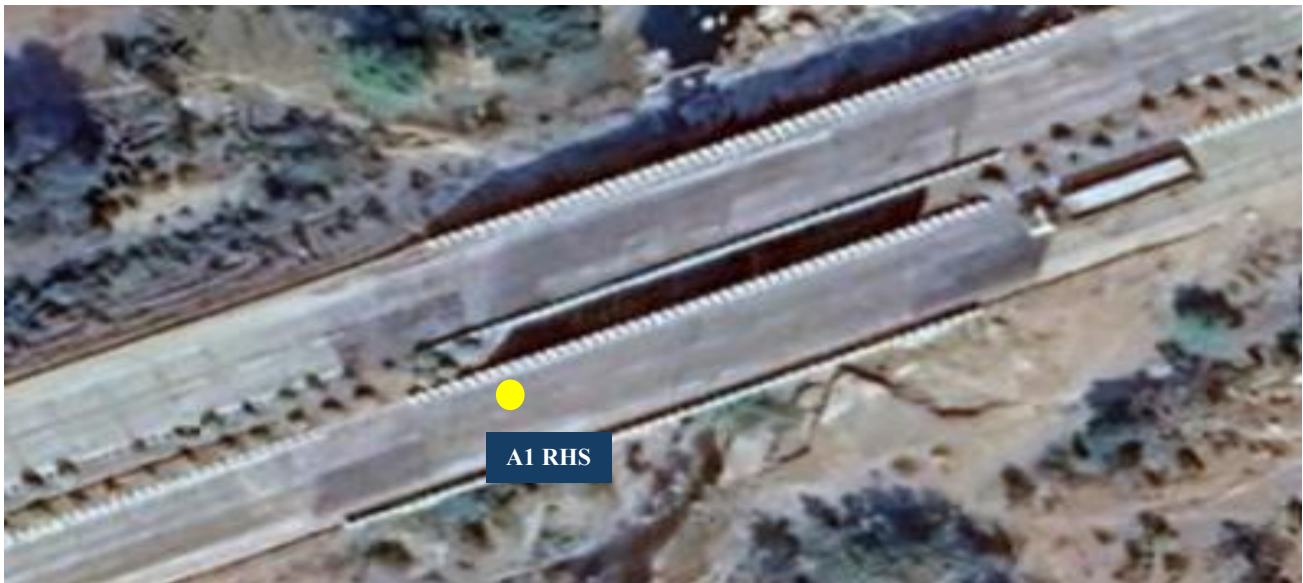
Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** The vertical displacement exhibits moderate fluctuation with an overall slightly upward tendency through the monitoring period.
- **Horizontal displacement (VH)** Horizontal displacement demonstrates greater variability compared to VV, fluctuating between approximately -5 mm and +12 mm.

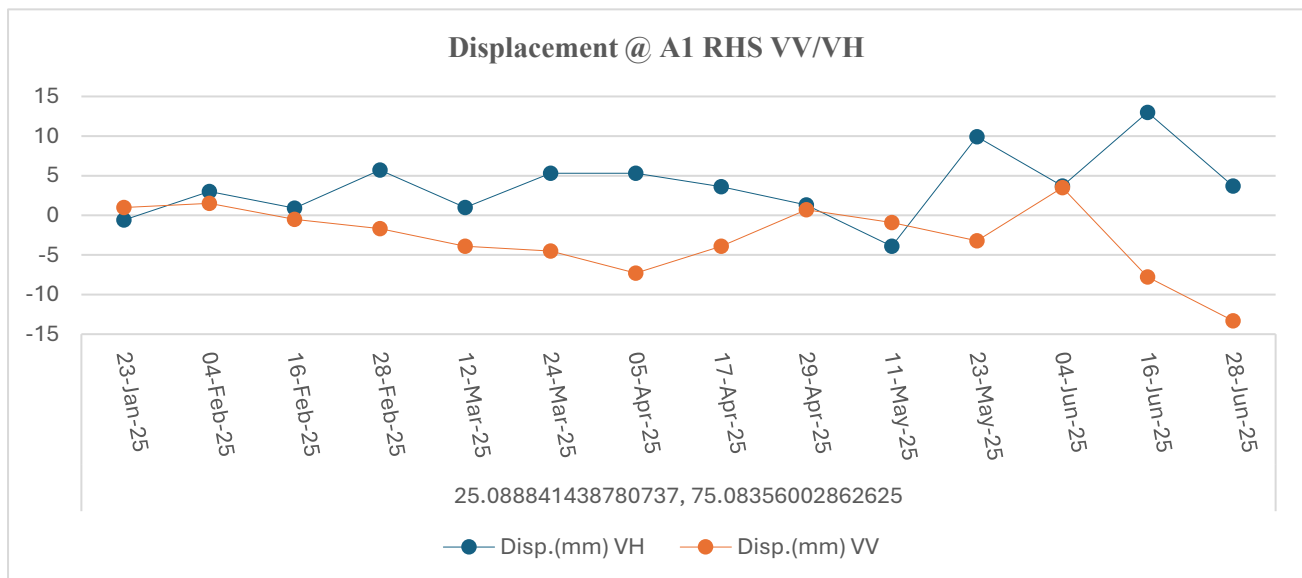
Critical Events & Anomalies:

- **January 2025:** VH records an early positive peak (around +9–10 mm), while VV remains mildly positive, marking an initial lateral movement event.
- **May 2025:** Another significant VH peak occurs (approximately +10–11 mm), accompanied by a modest VV increase, suggesting renewed lateral displacement activity
- **June 2025:** VV experiences a sharp downward excursion, reaching approximately -12 mm, the lowest vertical value in the record, while VH also declines, marking the most critical combined downward event.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.088841438780737, 75.08356002862625)



Key Observations:

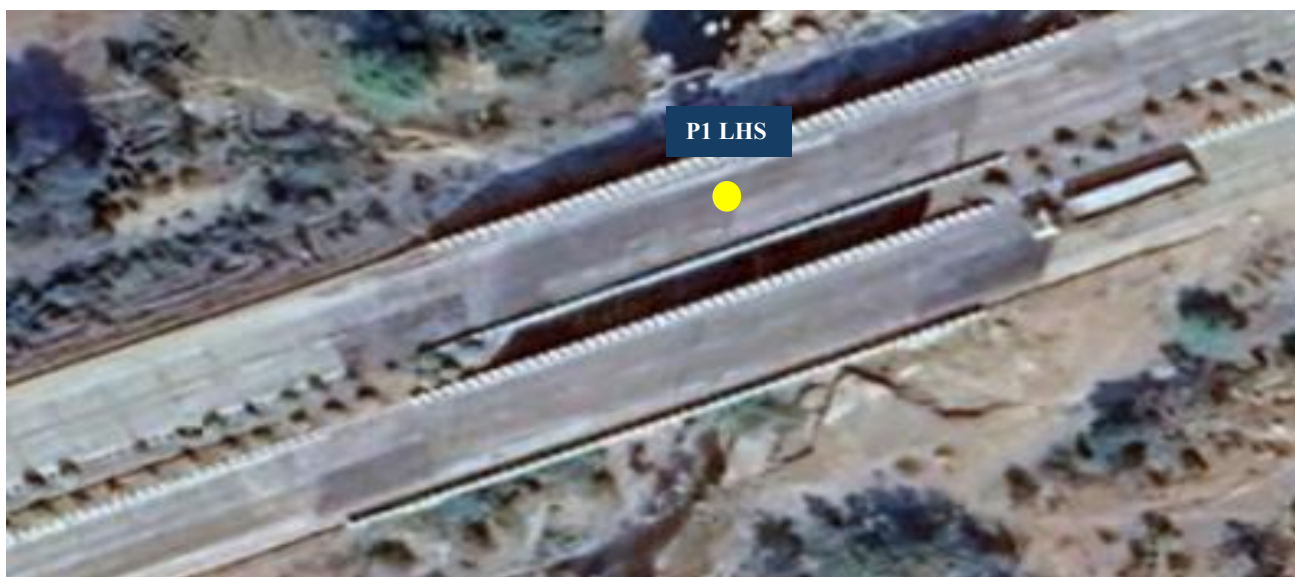
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement remains consistently active, fluctuating between approximately -4 mm and +13 mm.
- **Vertical displacement (VV)** Vertical displacement shows a predominantly downward trend over the monitoring period.

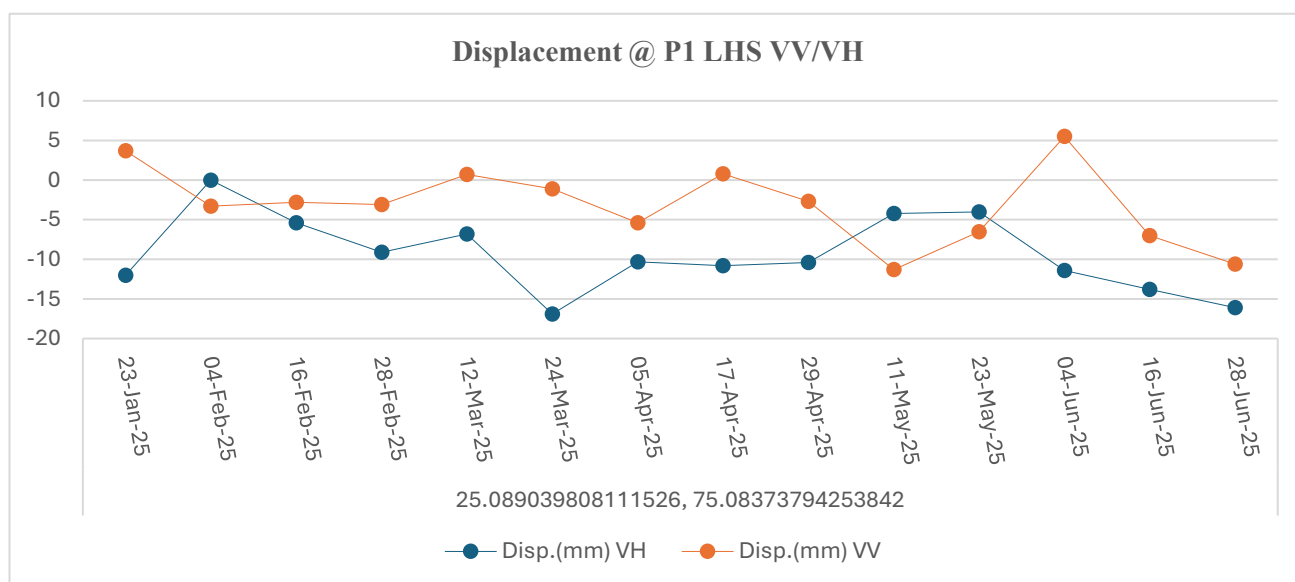
Critical Events & Anomalies:

- **Jan-Feb 2025:** VH and VV both remain close to zero with small positive values, indicating initial stability with minimal displacement.
- **March-April 2025:** VV undergoes a sustained downward trend, reaching approximately -7 mm, while VH stays moderately positive, suggesting settlement-dominated behaviour.
- **June 2025:** VH shows a strong positive peak (+9 mm) and VV briefly recovers to around +4 mm, suggesting a temporary uplift and lateral activation.

3.5 SAR Based Analysis at Pier 1 LHS:



P1 LHS (25.089039808111526, 75.08373794253842)



Key Observations:

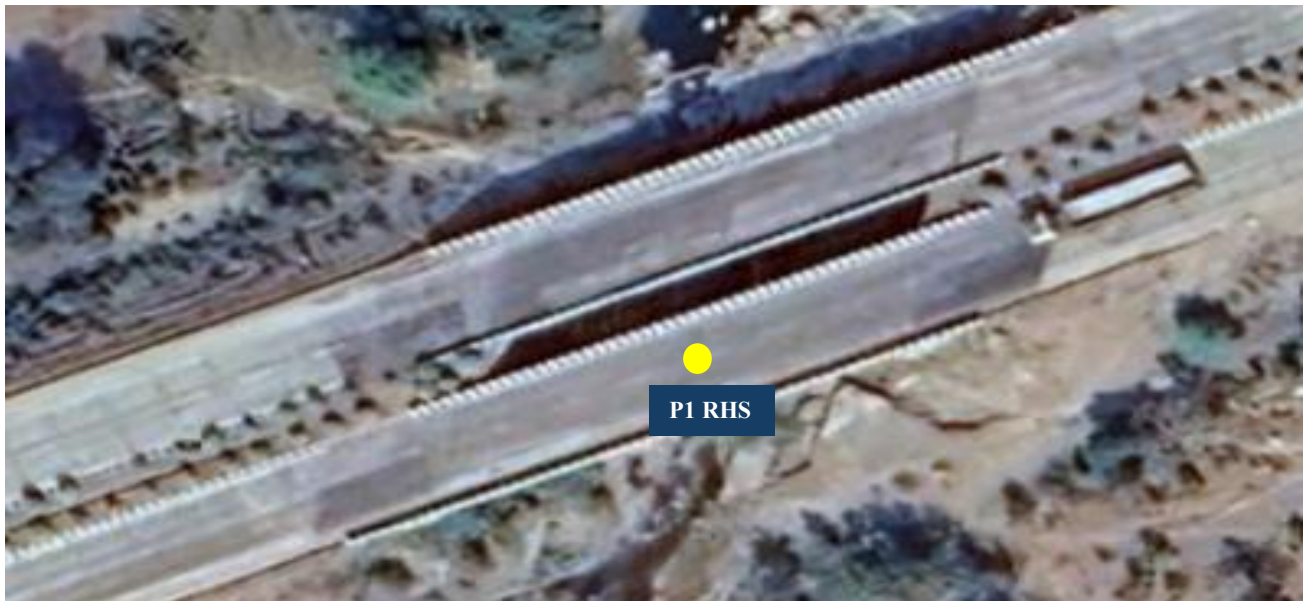
Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement exhibits alternating uplift and settlement throughout the monitoring period, without a sustained long-term trend.
- **Horizontal Displacement (VH)** Horizontal displacement shows a dominant negative trend, indicating persistent lateral movement in one direction.

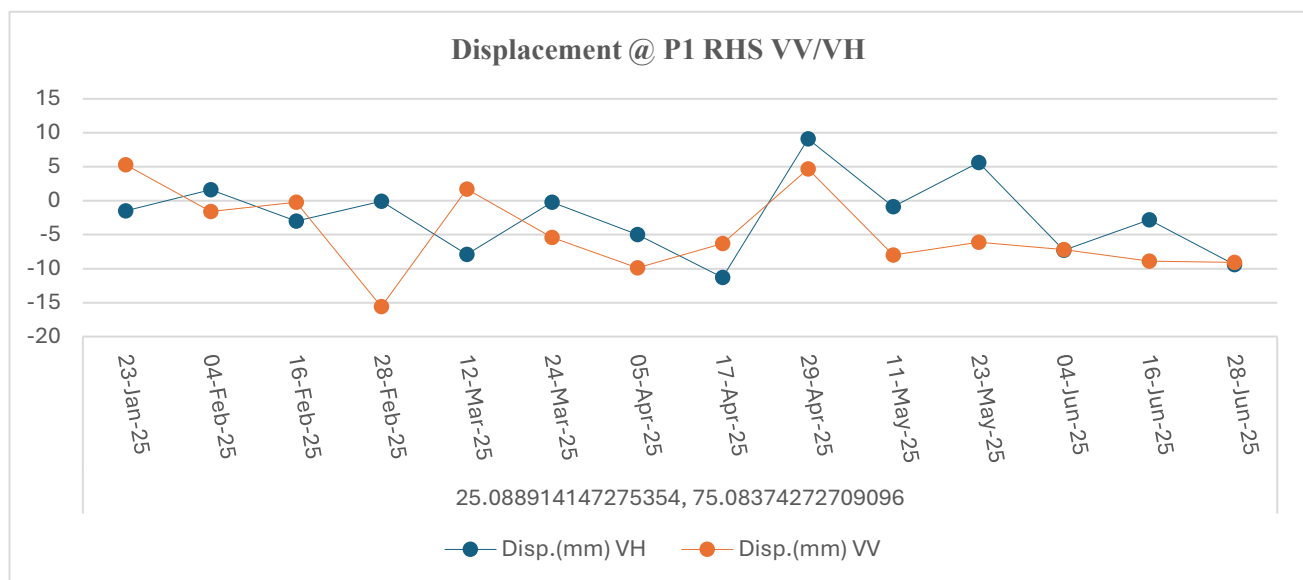
Critical Events & Anomalies:

- **January 2025:** VH starts at a relatively low value (around -12 mm), while VV is moderately positive, indicating initial lateral displacement with slight vertical uplift.
- **May 2025:** VH shows a partial improvement (around -5 mm), while VV reaches approximately -11 mm, indicating a short-term vertical settlement episode.
- **June 2025:** VH declines further to approximately -16 mm, while VV also trends downward, marking a **combined displacement event and the end-period critical condition**

3.6 SAR Based Analysis at Pier 1 RHS:



P1 RHS (25.088914147275354, 75.08374272709096)



Key Observations:

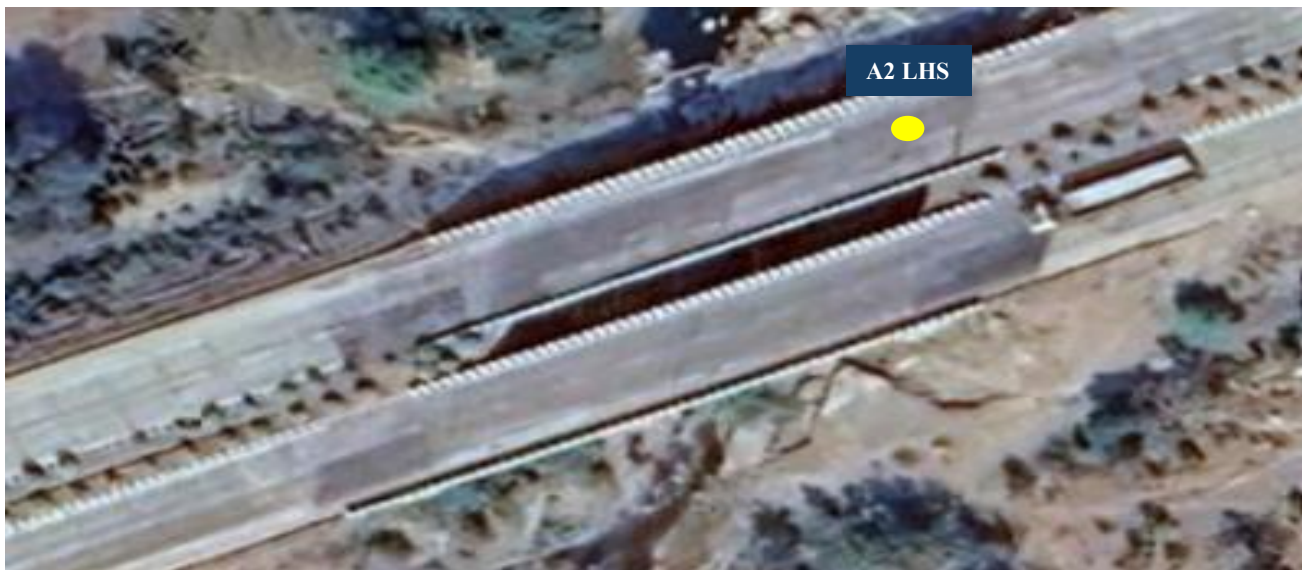
Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** The vertical response is predominantly negative (downward) after mid-March, indicating sustained settlement-type behaviour rather than recovery.
- **Horizontal Displacement (VH)** The displacement trend fluctuates throughout the monitoring period without a consistent directional bias, indicating reversible or cyclic lateral response rather than continuous drift.

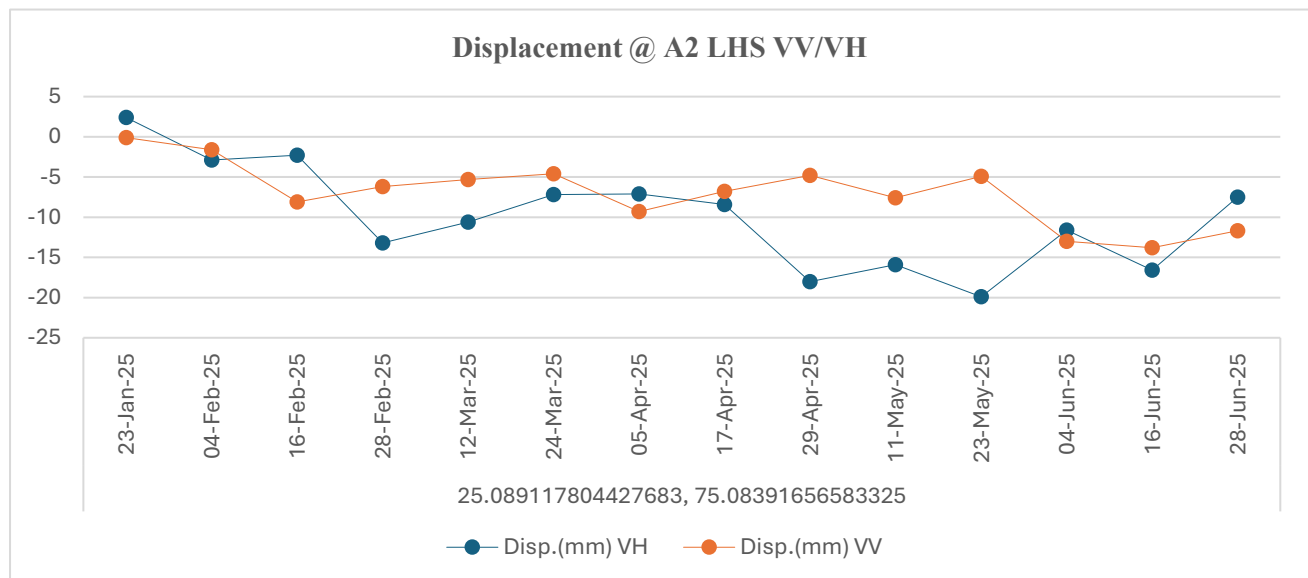
Critical Events & Anomalies:

- **Feb 2025:** VV experiences a sharp downward excursion (–15 mm), marking the largest vertical deviation in the monitoring period.
- **April 2025:** VH reaches its peak positive value (+8 mm) while VV transitions from negative to near-neutral, suggesting a short-term lateral adjustment or load redistribution.
- **June 2025:** Both VV and VH trend downward again (VV –10 mm, VH –5 mm), indicating renewed deformation activity rather than long-term stabilization.

3.9 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.089117804427683, 75.08391656583325)



Key Observations:

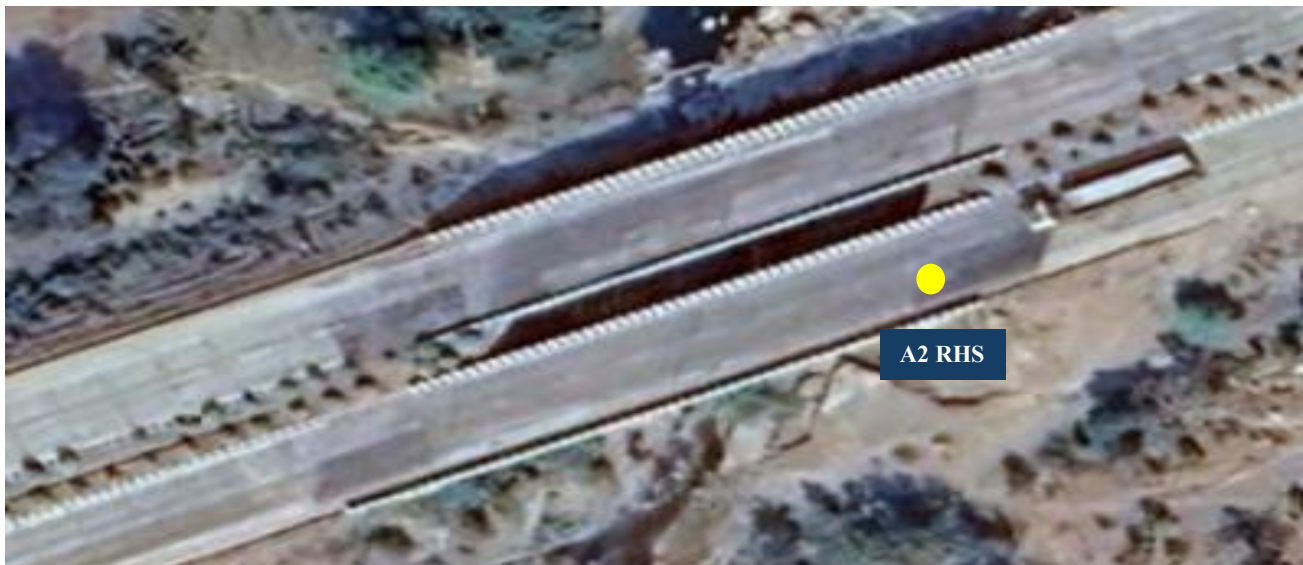
Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** VV remains consistently negative throughout the monitoring period, generally ranging between –3 mm and –15 mm, indicating sustained downward movement.
- **Horizontal Displacement (VH)** After early February, the horizontal movement trends predominantly negative, indicating lateral shift in a dominant direction. VH shows larger variability than VV, ranging approximately from +3 mm to –20 mm.

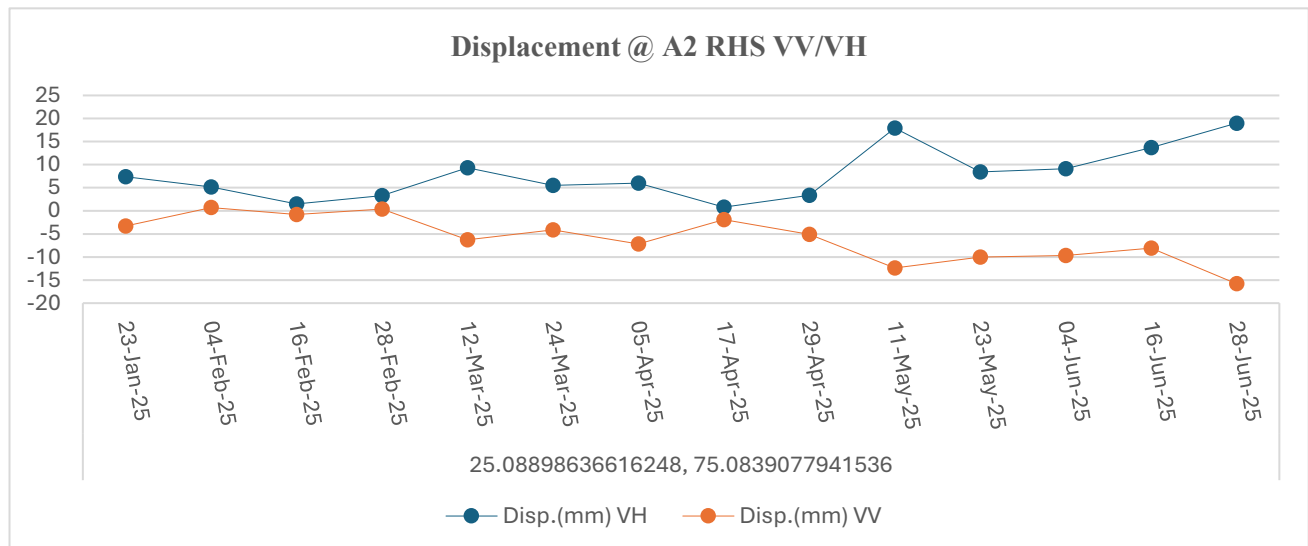
Critical Events & Anomalies:

- **February 2025:** VH drops sharply from near –2 mm to approximately –14 mm, representing the first major lateral displacement event.
- **June 2025:** Both VV (–12 to –15 mm) and VH (–12 to –17 mm) remain elevated in magnitude, indicating persistent deformation rather than stabilization.

3.10 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.08898636616248, 75.0839077941536)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** remains consistently negative throughout the monitoring period, ranging from approximately -2 mm to -16 mm, indicating sustained downward movement at A2 RHS.
- **Horizontal Displacement (VH)** VH exhibits a predominantly positive and increasing trend, ranging from approximately +2 mm to +19 mm. After mid-April, the horizontal displacement shows a clear escalation, indicating sustained lateral movement in one dominant direction

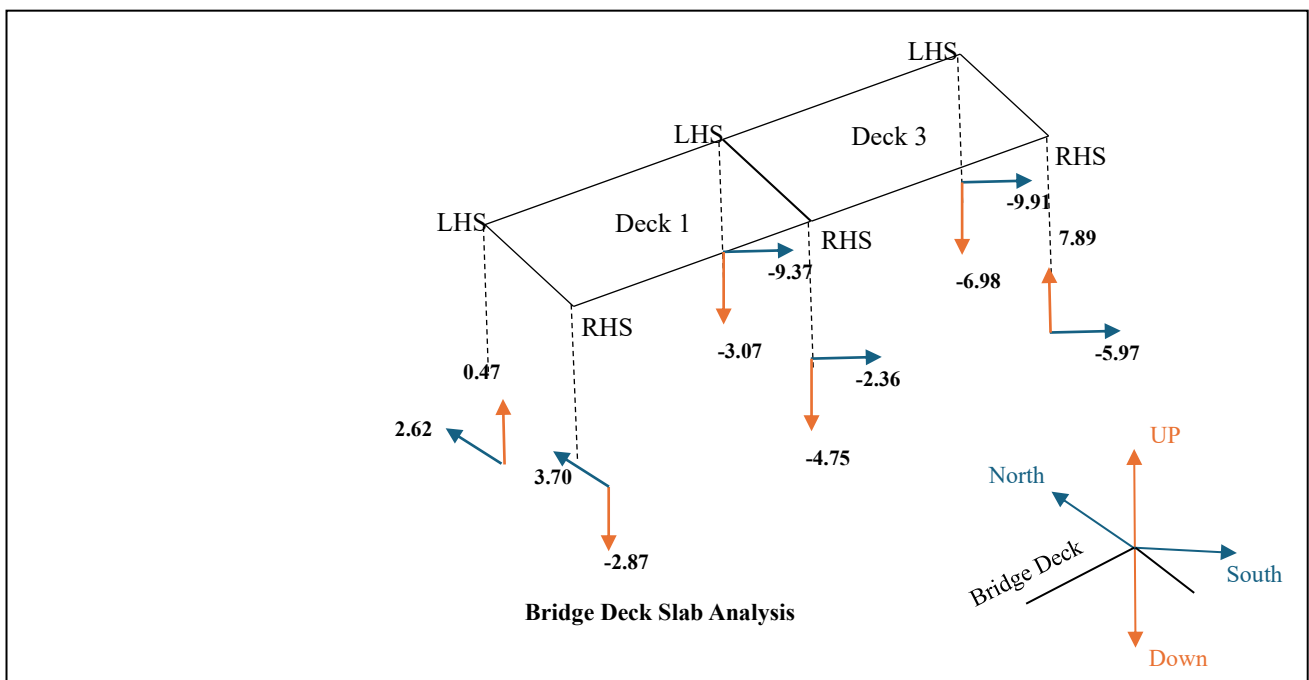
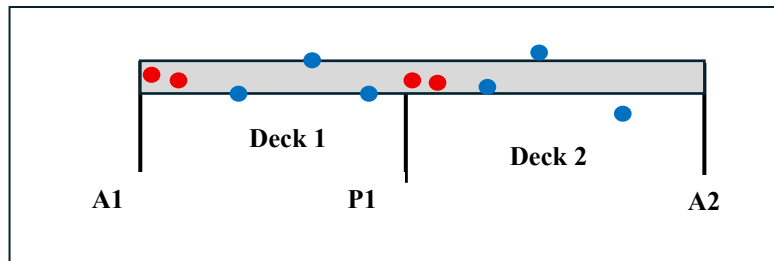
Critical Events & Anomalies:

- **March 2025:** VH rises sharply to around +10 mm, marking the first notable lateral shift. VV simultaneously drops to approximately -7 mm, indicating concurrent vertical settlement
- **May 2025:** VH reaches approximately +18 mm, the highest horizontal displacement recorded during the monitoring period. VV deepens to around -13 mm, suggesting coupled vertical-horizonal deformation
- **June 2025:** VH continues to increase, peaking near +19 mm, while VV reaches its maximum downward value (-16 mm). The absence of rebound indicates ongoing deformation rather than stabilization.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- Across Decks 1, 2, and 3, the recorded vertical displacements show small to moderate downward movements with limited uplift at selected locations, remaining generally within a narrow range (approximately -7 mm to +8 mm).

Horizontal Displacement Behaviour:

- Horizontal displacements across the three decks are relatively small, generally ranging between about 2 mm and 10 mm. Deck 1 shows limited lateral movement with no clear dominant direction, indicating effective restraint and stable bearing performance. Deck 2 displays modest lateral shifts (around 2–4 mm), which appear gradual and uniform, typical of thermal expansion–contraction of the superstructure. Deck 3 records slightly higher horizontal movement at the RHS (up to 10 mm), but without progressive increase or asymmetry that would suggest translational instability.

CONCLUSION:

- The combined vertical and horizontal displacement patterns across Decks 1, 2, and 3 are consistent with expected service-level behaviour of a bridge deck slab subjected to temperature variation, routine traffic loading, and seasonal effects. No abnormal settlement, excessive uplift, or systematic lateral drift is evident in the monitored data.